

RESEARCH EXPERIENCE

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| • Generative Models | 2023 - Present |
| • Video Generation; Diffusion Models | |
| • 3D Vision | 2023 - Present |
| • Gaussian Splatting; Point Cloud Segmentation; Stereo Matching | |
| • Multi-Modal Multi-Task Learning | 2023 - Present |
| • Medical Foundation Models; Modality Alignment | |
| • Object Detection in the Open World | 2020 - 2024 |
| • Domain Adaptation; Open-set Recognition; Open-Vocabulary Detection | |
| • Graph Learning | 2020 - 2024 |
| • Causality; Graph Convolution; Graph Matching; Probabilistic Graph | |
| • Medical Imaging Analysis | 2020 - 2024 |
| • MRI Brain Analysis; Tooth Analysis; Medical Report Generation | |

WORK EXPERIENCE

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|---|-----------------------|
| • The Chinese University of Hong Kong (CUHK) | Feb. 2024 - Dec. 2024 |
| • Postdoc, Electronic Engineering | |
| Supervisor: Prof. Yixuan Yuan | |

EDUCATION

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|---|-------------------------------|
| • City University of Hong Kong (CityU) | Aug. 2020 - Dec. 2023 |
| • Ph.D., Electrical Engineering | Early Graduation |
| Supervisor: Prof. Yixuan Yuan | |
| • Tianjin University (TJU) | Sep. 2016 - Jun. 2020 |
| • B.S., Electrical Engineering | GPA: 3.83/4.0 (Top 2%) |

SELECTED CONFERENCE PAPER

1. **CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection**
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
 European Conference on Computer Vision (**ECCV**), 2024
2. **Boosting Medical Image Analysis via Motion-Informed Generative Videos**
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
3. **Novel Scenes & Classes: Towards Adaptive Open-set Object Detection**
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
 IEEE International Conference on Computer Vision (**ICCV**), 2023, **Oral (1.8%)**
4. **Adjustment and Alignment for Unbiased Open Set Domain Adaptation**
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023
5. **SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022, **Best Paper Finalist (0.4%)**
6. **SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation**
Wuyang Li, Xinyu Liu, Xiwen, Yao, Yixuan Yuan
 The Association for the Advance of Artificial Intelligence (**AAAI**), 2022, **Oral (4.5%)**
7. **GaussianStego: A Generalizable Stenography Pipeline for Generative 3D Gaussians Splatting**
 Chenxin Li, Hengyu Liu, Zhiwen Fan, **Wuyang Li**, Yifan Liu, Panwang Pan, Yixuan Yuan
 Under Review, NeurIPS, 2024
8. **When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels**
 Yifan Liu, **Wuyang Li**, Cheng Wang, Hui Chen, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024, **Early Accept (14.0%)**

9. **DiffRect: Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation**
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention ([MICCAI](#)), 2024
10. **F2TNet: FMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction**
Zhibin He, Wuyang Li, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
Medical Image Computing and Computer Assisted Intervention ([MICCAI](#)), 2024
11. **LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction**
Hengyu Liu*, Yifan Liu*, Chenxin Li*, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention ([MICCAI](#)), 2024
12. **Endora: Video Generation Models as Endoscopy Simulators**
Chenxin Li, Hengyu Liu, Yifan Liu, Brandom Feng, Wuyang Li, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention ([MICCAI](#)), 2024
13. **MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics**
Qiushi Yang, Wuyang Li, Baopu Li, Yixuan Yuan
IEEE International Conference on Computer Vision ([ICCV](#)), 2023
14. **H2GM: A Hierarchical Hypergraph Matching Framework for Brain Landmark Alignment**
Zhibin He, Wuyang Li, Tuo Zhang, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention ([MICCAI](#)), 2023
15. **Towards Robust Adaptive Object Detection under Noisy Annotations**
Xinyu Liu, Wuyang Li, Qiushi Yang, Baopu Li, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition ([CVPR](#)), 2022
16. **Intervention & Interaction Federated Abnormality Detection with Noisy Clients**
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention ([MICCAI](#)), 2022, **Oral**

SELECTED JOURNAL PAPER

1. **SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
IEEE Transactions on Pattern Analysis and Machine Intelligence ([TPAMI](#)), 2023
2. **HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection**
Wuyang Li, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
IEEE Transactions on Image Processing ([TIP](#)), 2021
3. **Stereo Vision Meta-Lens-Assisted Driving Vision**
Xiaoyuan Liu, Wuyang Li, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
ACS Photonics, 2024, **Selected as Cover Paper**
4. **LLM-guided Decoupled Probabilistic Prompt for Continual Learning in Medical Image Diagnosis**
Yiwen Luo, Wuyang Li, Chen Cheng, Xiang Li, Tianming Liu, Yixuan Yuan
IEEE Transactions on Medical Imaging ([TMI](#)), 2024, Under Revision
5. **FM-APP: Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer**
Zhibin He, Wuyang Li, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
IEEE Transactions on Medical Imaging ([TMI](#)), 2024, Under Review
6. **Medical Federated Learning with Joint Graph Purification for Noisy Label Learning**
Zhen Chen, Wuyang Li, Xiaohan Xing, Yixuan Yuan
Medical Image Analysis ([MIA](#)), 2023
7. **Decoupled Unbiased Teacher for Source-free Domain Adaptive Medical Object Detection**
Xinyu Liu, Wuyang Li, Yixuan Yuan
IEEE Transactions on Neural Networks and Learning Systems ([TNNLS](#)), 2023
8. **GRAB-Net: Graph-based Boundary-aware Network for Medical Point Cloud Segmentation**
Yifan Liu, Wuyang Li, Jie Liu, Hui Chen, Yixuan Yuan
IEEE Transactions on Medical Imaging ([TMI](#)), 2023

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1. **U-KAN Makes Strong Backbone for Medical Image Segmentation and Generation**
Chenxin Li*, Xinyu Liu*, Wuyang Li*, Cheng Wang*, Hengyu Liu, Yixuan Yuan (* Equal Contribution)
ArXiv preprint arXiv:2406.02918, 2024

SELECTED AWARDS

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- **Outstanding Academic Performance Award, CityU** 2022 - 2023
Outstanding research achievements
 - **Research Tuition Scholarship, CityU** 2022 - 2023
Outstanding research achievements
 - **Ranked First in Polyp Detection Challenge** 2021
3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021)
 - **National Scholarship** 2017 - 2018
The great honor for a student in China
 - **Silver Award in Tianjin Advanced Mathematical Competition** 2017
As a selected participant representing Tianjin University

SERVICES

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- **NeruIPS-24 Workshop Proposal [Link]** 2024
Assisted the *Advancements in Multi-modal Medical Foundation Models* workshop [Proposal Link]
 - **Research Funding Proposal** 2022 - 2024
Assisted my advisor in completing five research funding proposals over the past two years
 - **Startup Funding Proposal [Link]** 2023
Successfully secured seed funding from HK Tech 300 and HKSTP
 - **Founder of ScholaGO Education Technology Company Limited, Hong Kong [Link]** 2024
Founded a company to develop a multi-modal educational product
 - **Conference and Journal Reviewer**
T-PAMI; IJCV; TIP; TNNLS; T-IV; T-ASE; PR; NeurIPS-24; CVPR-23/24; ICCV-23; AAAI-23/24